

Gap Schemes: Power, Counter, Duo, Trap, Wham, Iso, and the Draw

Why a guard runs sideways at the snap, and how an offense brings more blockers to one hole than the defense has defenders there

Third-and-1, the offense in a heavy formation with a fullback and a tight end, everyone in the stadium knowing it will run. At the snap the right side of the offensive line fires *inward*, at an angle, away from where the ball is going. At the same instant the left guard, who should be blocking the man in front of him, turns and sprints sideways behind his own line. The fullback heads for the edge. The running back waits a beat, then follows the guard into a hole just outside the tight end, and the guard turns up into it and meets a linebacker head-on. Four yards, first down. The announcer says the offense “just lined up and ran it right at them.” True, but not the interesting part. Everyone knew where the ball was going, so how did the offense get more blockers to that spot than the defense had defenders? The answer is a family of plays coaches call **gap schemes**, and the guard who ran sideways is the key to every one of them.

You’ll be able to:

- Explain the three parts of every gap run (down blocks, a puller or two, and a kick-out) and why each exists.
- Recognize power, counter, duo, trap, wham, iso, the toss and sweep, pin-and-pull, and the draw on a broadcast, and draw each against a front you are shown.
- Count the blockers and defenders at the point of attack and show how a gap scheme gains a man there, and what it pays for him.
- Say when a coach is likely to call a gap run rather than a zone run (the front, the personnel, the offensive line, the situation).
- Spot the pre-snap and first-step tells of a pulling lineman, and call the play’s direction from them.

You’ll need

- **The Language of the Run Game:** gaps A to D, technique numbers (a **3-technique** on a guard’s outside shoulder, a **5** on a tackle’s outside shoulder, a **1** shaded on the center), covered and uncovered linemen, play side and backside, the point of attack, hat math, and the eight core blocks, especially the **down block**, the **pull**, the **kick-out** and the **lead block**. This chapter is built from those words.
- Helpful but not required: Zone Running, the other great family of runs, which this chapter keeps comparing against; the I-formation and the fullback; and the stance tells from How to Watch a Broadcast.

How to read the diagrams. Offense is blue circles, defense orange squares, and the offense moves up the page. *Left* and *right* are always the offense’s left and right. LT, LG, C, RG and RT are the five offensive linemen; Q is the quarterback, R the running back, F a fullback, H an H-back or move tight end, and Y and U are tight ends on the right and left. Defensive linemen carry their technique number; S, M and W are the Sam, Mike and Will linebackers; FS and SS the safeties; CB a cornerback; NB the nickel back. Lines ending in a short bar are blocks; plain arrows are paths; the thick dark arrow is the ball carrier. **A purple ring marks a puller**, the player this chapter wants you to follow. Most diagrams zoom in on the box, so the wide receivers (who keep the formation legal, with seven men on the line) are often outside the picture.

What makes a run a “gap” run?

In Zone Running the offensive line asked one question after the snap: *who comes into my area?* A gap scheme asks its question *before* the snap: *which man do I block, and from which angle?* That is why gap blocking is also called **man** or **power** blocking. A **gap scheme** (also **man/power blocking**) assigns each blocker a man or a gap in advance, picks those assignments to give blockers good angles, and brings extra blockers to a hole the coach chose before the snap.

Every gap run, from the 1960s Packers to the 2024 Eagles, is assembled from three parts you met in 03-01:

1. **Down blocks on the play side.** The linemen at the point of attack block the defender *inside* them, hitting him from the side. A down block is the easiest block in football to win, because the blocker has an angle and the defender is looking straight ahead. Down blocks build the inside wall of the hole.
2. **A kick-out on the end man.** Somebody (a fullback, an H-back, a pulling guard) hits the end man, the last defender on the line, from the inside and drives him toward the sideline. That builds the outside wall.

3. **A puller through the hole.** A lineman from the *backside* runs behind his own line, turns up between the two walls and blocks the first linebacker to show. He is the extra man.

Down blocks in, kick-out out, puller up through the middle: that is the whole idea. The hole is fixed and known to everyone on the offense, including the running back, who does not have to find it; he has to *follow* someone into it. Contrast zone, where the hole is wherever the wall cracks and the back's read is half the play.

The lineman who leaves his spot is the **puller**: the guard (or tackle, or both) assigned to pull on this play. He is the most important player on the field for you to find, because **a puller runs to the hole**. The rest of this chapter is, in a sense, a catalogue of where pullers go.

! Why it exists

Gap schemes exist to **win the numbers at the point of attack**. A defense can always put as many men in the box as the offense has blockers, and usually one more (hat math). It cannot be equally strong everywhere, though, and it has to line up before it knows where the ball is going. A gap scheme picks one spot, then moves blockers there from elsewhere (a backside guard, a fullback) faster than the defense can move defenders there. It pays for those men by leaving someone *away* from the play unblocked, and by being more predictable: a pulling guard is a signpost. Every gap play in this chapter is a different answer to the same two questions: *where do the extra blockers come from, and who is left free?*

Power: the play the whole family is built on

Power (often called **Power O**) is the purest gap run, and the one to learn first. Here it is from the I-formation (a fullback behind the quarterback and the tailback behind him) against a **4-3 Over**, a four-man front whose 3-technique lines up on the side of the tight end (fronts are named and explained in Even Fronts; here it is just a set of numbers to block). Read it from the right, the play side, to the left.

Walk through Figure 1 one man at a time, because each assignment exists for a reason.

The play-side wall: down blocks. The tight end blocks *down* on the 5-technique inside him, and the right tackle blocks down on the 3-technique inside *him*. Neither has to beat a defender lined up across from him; each hits a man from the side, which is why gap teams love the down block. The right guard joins the tackle on the 3-technique for a moment, a combo block, and then climbs to the backside linebacker, the Will. Coaches teach these linemen one short rule, usually some version of *gap, down, backer*: if a defender is in your inside gap, block him; if not, block down on the man over the lineman inside you; if nobody is there, go to a linebacker.¹ The rule means a play-side lineman never has to reach a man lined up outside him.

The center blocks back. The left guard is about to leave. The defensive lineman who was across from the gap he leaves, here the 1-technique, would walk into the backfield, so the center blocks *back* on him, away from the play. Because the 1 is shaded on the center's backside shoulder, it is another easy angle. This is not a coincidence: coaches like to call power *toward* a 3-technique and *away* from a 1-technique for exactly these angles.

¹"Gap, down, backer" (sometimes "gap, down, linebacker" or "GDL") is the standard play-side rule for power and counter in coaching manuals and clinic talks. One coaching write-up states it as: gap, "if there is somebody in your immediate backside gap, block them"; down, "if there is somebody over or inside shoulder of the linemen in your backside gap, block them"; backer, "if there is nobody in the gap or down, head to the linebacker": Chris Haddad, "How To Install Gap Scheme Runs Into Your Offense," Viqtory Sports, <https://www.viqtorysports.com/the-gap-scheme-how-its-run-in-every-offense/>. For power's assignments in general, see Chris B. Brown, *The Art of Smart Football* (2015).

Power right vs a 4-3 Over

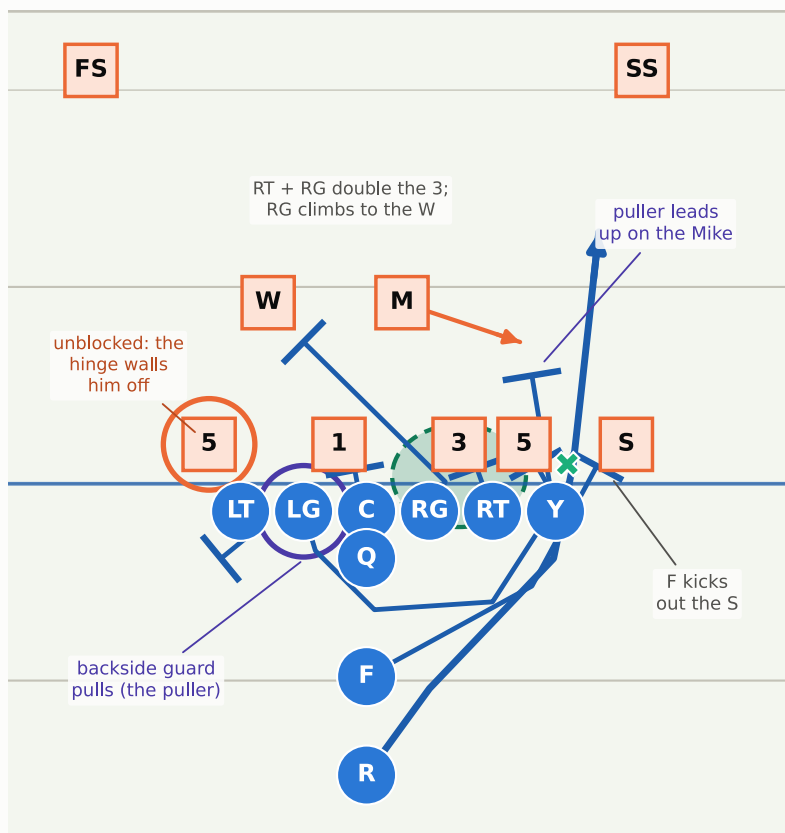


Figure 1: Power right against a 4-3 Over. Play side: the Y blocks down on the 5-technique, and the RT and RG double-team the 3-technique (green oval) until the RG climbs to the Will. The center blocks back on the 1. The fullback kicks out the end man, the Sam standing up in a 9. The left guard (purple ring) pulls behind the line, turns up through the hole (green X) between the Y's down block and the kick-out, and leads on the Mike as he scrapes toward the hole. The left tackle hinges: he steps inside to close the gap the puller left, then turns out. The backside end (orange ring) is the one defender nobody blocks, and he is the farthest from the ball.

The kick-out. The fullback runs at the end man (here the Sam, standing up on the line outside the tight end in a 9-technique) and hits him from the inside, driving him toward the sideline. The fullback does not need to win big. If the Sam stays outside, the hole is inside him.

The puller. The left guard opens, sprints flat behind the center and the right side, and turns up into the hole, just outside the tight end's down block and inside the kick-out. His man is the first linebacker who shows, almost always the play-side linebacker (the Mike here) coming to fill. The running back follows him, aiming his first steps at the tight end's outside hip and staying a step behind the guard's back, then cutting off the guard's block.

The backside. With the guard gone, the left tackle **hinges**: he steps inside to close the B gap the guard left, then swings back outside like a door closing, to keep the backside end from running straight down the line. He does not try to drive the end anywhere. The end is the one defender power *chooses* not to block. That is the deal at the heart of the play.

Where power's extra blockers come from

Count the bodies, as in hat math, and power's logic becomes arithmetic.

Where power's extra blockers come from

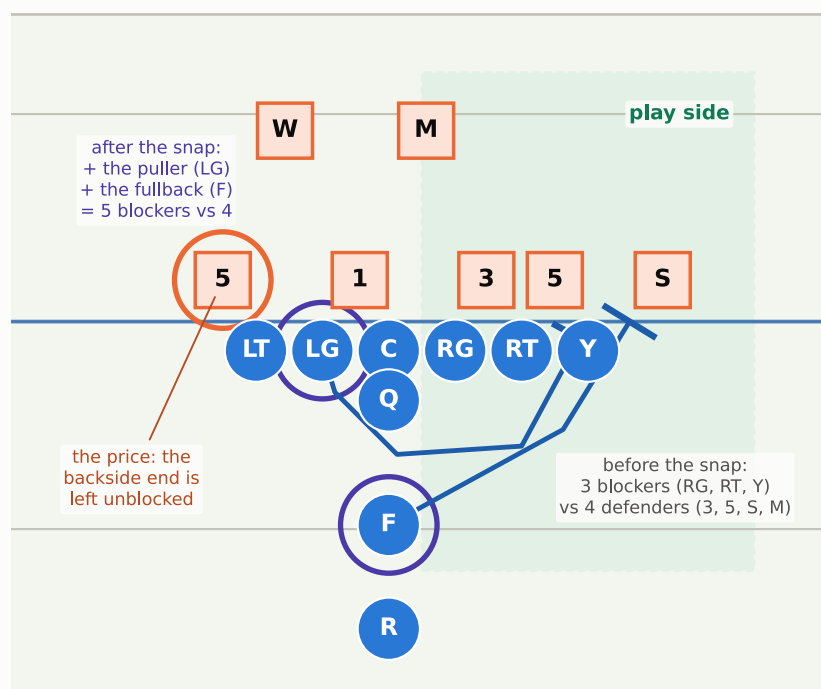


Figure 2: Power's arithmetic, against the same 4-3 Over. Before the snap the play side (shaded) holds three blockers (RG, RT, Y) and four defenders (the 3, the 5, the Sam and the Mike): the offense is a man short where it wants to run. At the snap two more blockers arrive from the backside, the pulling left guard and the fullback (purple rings), so the play side is five on four. The offense paid for them by not blocking the backside end (orange ring), the defender farthest from the hole.

Figure 2 is the reason gap schemes exist. Before the snap, the side the offense wants to run to has three blockers and four defenders. Run zone that way and somebody has to beat a man he is outnumbered against. Run power and two blockers who started on the *other* side of the ball arrive at the point of attack within a second: five on four. The defense cannot match that move, because its players are reading the snap, not leading it.

The bill comes due on the backside. The offense moved two hats across, so the backside is short: the center, the hinging left tackle and nobody else against the 1, the Will and the backside end. The right guard's climb takes care of the Will, the center takes the 1, and the end is left alone. Power is betting that a defender who lines up farthest from the hole, and has to run down the line behind a wall of bodies, will arrive too late. Most of the time he does. When the defense *knows* that and sends that end screaming down the line at the snap, the offense has answers (a quarterback who keeps the ball and runs where the end left, in Option Football; a bootleg pass behind him, in Play-Action, Boots and Screens).

Common misconception

"Power works because the offensive line is bigger and stronger." Strength helps, but the play is designed so that nobody has to win a fair fight. The down blocks hit men from the side; the kick-out hits the end man from inside, where he is not looking; the puller arrives as an extra man. What power needs is not the biggest line but a line that can *pull*: a guard who can run ten yards laterally and still hit a linebacker. That is why gap-scheme teams prize athletic guards as much as heavy ones.

The name, by the way, tells you less than you would hope. "Power" describes the play's attitude (downhill, a hat on every hat and one more) rather than any single assignment, and the "O" in Power O is not explained consistently from one coaching tree to the next. Teams also run power from the shotgun, with an H-back or a second tight end as the kick-out man instead of a fullback; the structure does not change.

Counter: the same blocks, a different first step

Power has one weakness that defenses learn quickly: the running back's first step points at the hole, and so does the puller. A linebacker who reads both is moving toward the hole before the guard gets there. **Counter** attacks that reflex. It starts with the back stepping the *wrong* way, and it pulls **two** backside blockers the right way: classically the guard, who kicks out the end man, and the tackle, who wraps up through the hole as the lead blocker. The guard-and-tackle version is called **GT counter**, and its most famous name is the **counter trey** of Joe Gibbs's Washington teams.

Look at what changed from power in Figure 3, and what did not. The play side is identical: the right tackle blocks down on the 3-technique, the right guard climbs to the backside linebacker, the center blocks back on the 1. What changed is *who supplies the extra men*. Instead of a fullback for the kick-out, the backside guard does it; instead of the guard leading through the hole, the backside tackle does. Two linemen leave the backside, so something has to fill the space they left: here the U tight end blocks the backside end, and in other versions an H-back fills, or the end is simply left for the quarterback to read.

And the running back's first step goes *away*. That counter step, a single lateral jab in the wrong direction, is the whole trick. Linebackers are taught to read the back and the guards. On counter the back says *left* while the guards say *right*, and a linebacker who trusts his eyes takes one false step toward the back, which is all the pulling tackle needs to arrive first. In the simulation the Mike's first two steps go toward the back's jab (frame 2), and by the time he corrects, the tackle is already in the hole.

Counter has a price too. Two pullers take longer to arrive than one, so the back's counter step is partly a way to buy time: the ball cannot hit the hole before the blockers do. If a defensive lineman penetrates into the space the pullers must run through, he can wreck both pullers at once. And a defense that keys the *guards* and ignores the back is not fooled at all, which is why offenses mix counter with power, zone, and fakes of all of them.

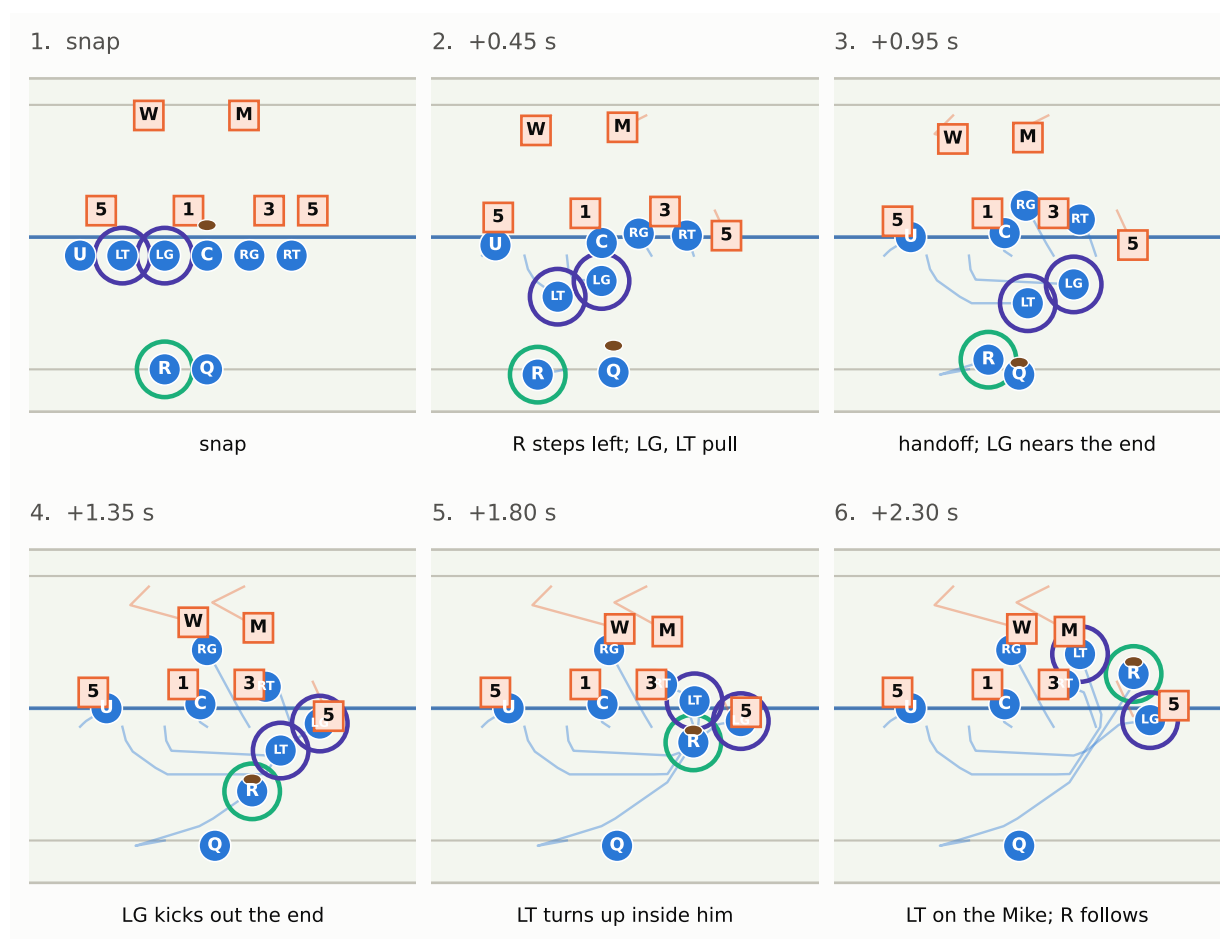


Figure 3: GT counter right from the shotgun against an even front (simulated play in Big Data Bowl coordinates; faint lines are each player's path so far). 1: the U tight end is on the left, the back offset left. 2: the back steps left, away from the play, as both the left guard and left tackle (purple rings) pull right; the right side blocks down and the center blocks back on the 1. 3-4: the handoff, and the guard reaches the end man and kicks him out. 5: the tackle turns up inside the kick-out. 6: the tackle leads up on the Mike, who had taken a false step toward the back's counter step, and the back (green ring) follows him through the hole. The U fills for the pulled tackle by blocking the backside end.

i Film room: Washington, 1983 season

The play: Gibbs installed the counter trey in his first season, 1981, and ran it for a decade. The Washington version pulled the left guard and left tackle (Russ Grimm and Joe Jacoby, two of the original Hogs) to the right, while the right side blocked down as if the play were going left and the back jabbed left before taking the handoff going right. Gibbs himself credited Tom Osborne's Nebraska teams: his staff watched Nebraska film, liked what Osborne's line was doing, "so, we stole it." (Nebraska's counter is usually dated to its 1982 season, a year after Gibbs arrived, so accounts of exactly which came first differ.)²

The season: in 1983, behind that line, 34-year-old John Riggins ran for 1,347 yards and 24 rushing touchdowns, then the NFL record, for a team that went 14–2.³

What to notice: on any Washington run from those seasons, ignore the ball and watch numbers 68 and 66 on the left side. If both of them pull right, it is the counter trey, and you can call the hole (just outside the right tackle's down block) before the back even turns back toward it.

Counter comes in many flavors that keep the same skeleton. Some teams pull the guard and an H-back or tight end instead of the tackle; some run it from the shotgun with the quarterback reading the backside end instead of blocking him (Option Football covers those reads); some run it *as* the quarterback's own run. When you see two linemen pulling the same way, think counter first.

New block: the log

Every kick-out has an assumption hidden in it: that the end man will stay outside and fight to keep his outside arm free. Many defenses teach exactly the opposite. The end is told to *squeeze*: to crash hard inside at the first sign of a down block, wrong-shoulder the puller and plug the hole, forcing the ball to bounce outside where his teammates are waiting. (Defenses call those two plans *spill* and *box*, and choosing between them is a big part of Run Fits and Box Math.) If the end has already crashed inside, the puller cannot kick him out: there is no "out" left to kick him toward.

So the puller adjusts on the run. If the end stays wide, he kicks him out and the ball goes inside. If the end squeezes inside, the puller throws a **log block**: he aims for the defender's *outside* shoulder, gets his head and body outside him and seals him in, and the ball carrier bounces outside the block. Some line coaches call this *wrapping* or *logging* the end; others use **wrap** for any puller who swings around a block in front of him and turns up, like the counter tackle above. Either way, the idea is the same: **the defender decides where the hole is, and the puller and the back read him.**

²"Counter run," Wikipedia (Washington's counter trey of the 1980s and early 1990s, "modeled from the Nebraska Cornhuskers"; the left guard and left tackle pull while the right side blocks down; the back steps left, then takes the ball right): https://en.wikipedia.org/wiki/Counter_run. Gibbs on Nebraska ("We saw some film on Nebraska, and Tom Osborne was doing some really innovative things with his line up front.... So, we stole it."), and Nebraska's counter dated to 1982: Dave Feit, "Greatest Huskers by the Numbers: 6," Sports Illustrated (All Huskers), <https://www.si.com/college/nebraska/football/dave-feit-s-greatest-huskers-by-the-numbers-6-keith-jones-darin-erstad-tom-osborne-national-champions>. The counter trey as "the Redskins' signature running play that Gibbs implemented his first season," with Grimm and Jacoby pulling: Associated Press, "Gibbs expected to use same bag of tricks" (2004), <https://www.nbcnews.com/id/wbna5946857>; Rick Gosselin, "State Your Case: Joe Jacoby": <https://rickgosselin.com/state-your-case-joe-jacoby/>. Jersey numbers (Grimm 68, Jacoby 66): Pro Football Hall of Fame, <https://www.profootballhof.com/players/russ-grimm/>; Hogs Haven, "Redskins by the (Jersey) Numbers: #66 - Joe Jacoby," <https://www.hogshaven.com/2017/8/9/16105032/redskins-by-the-jersey-numbers-66-joe-jacoby>. Washington's 14–2 record in 1983: "1983 Washington Redskins season," Wikipedia, https://en.wikipedia.org/wiki/1983_Washington_Redskins_season.

³John Riggins, 1983: 1,347 rushing yards and 24 rushing touchdowns, then an NFL single-season record (broken in 1995): Pro Football Hall of Fame, <https://www.profootballhof.com/players/john-riggins/>; "John Riggins," Wikipedia, https://en.wikipedia.org/wiki/John_Riggins; Pro Football Reference, <https://www.pro-football-reference.com/players/R/RiggJo00.htm>.

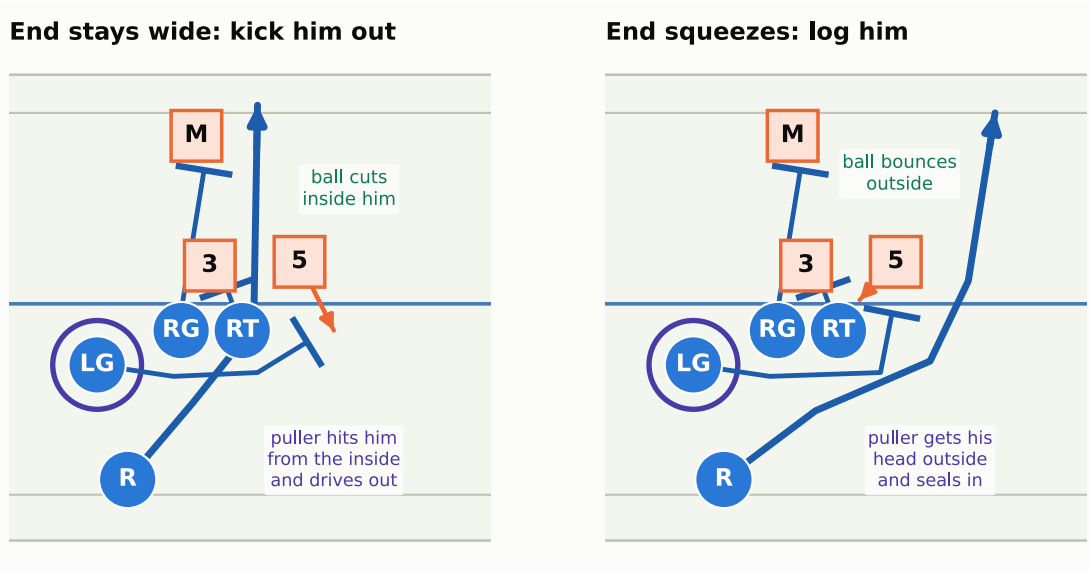


Figure 4: Kick or log: the puller (purple ring) reads the end man as he arrives. Left: the 5-technique comes upfield and stays outside, so the puller hits him from the inside and drives him out (a kick-out), and the back cuts inside him. Right: the end squeezes hard inside as the tackle blocks down (short orange arrow), so the puller swings to his outside shoulder and seals him in (a log block), and the back bounces outside. Same play, same puller; the defender chose the hole.

This is why gap runs are not as rigid as they look on a whiteboard. The *assignments* are fixed before the snap; the *hole* still moves a yard or two depending on what the end man does, and the back's eyes are on the puller's helmet. Where the helmet goes (inside or outside the defender), the back goes the other way.

Duo: power without a puller

Not every gap run pulls anybody. **Duo** is the downhill, no-frills member of the family: two double teams (hence the name) drive the two defensive tackles straight back, every other blocker takes the man in front of him, and nobody moves sideways. It looks like inside zone for the first instant, which is part of the point, but the footwork is different: zone linemen step *sideways* toward the play; duo linemen step *forward*, into the defenders, and the double teams stay on the defensive tackles longer, trying to move them back a yard or two before anyone climbs.

Duo vs two-high

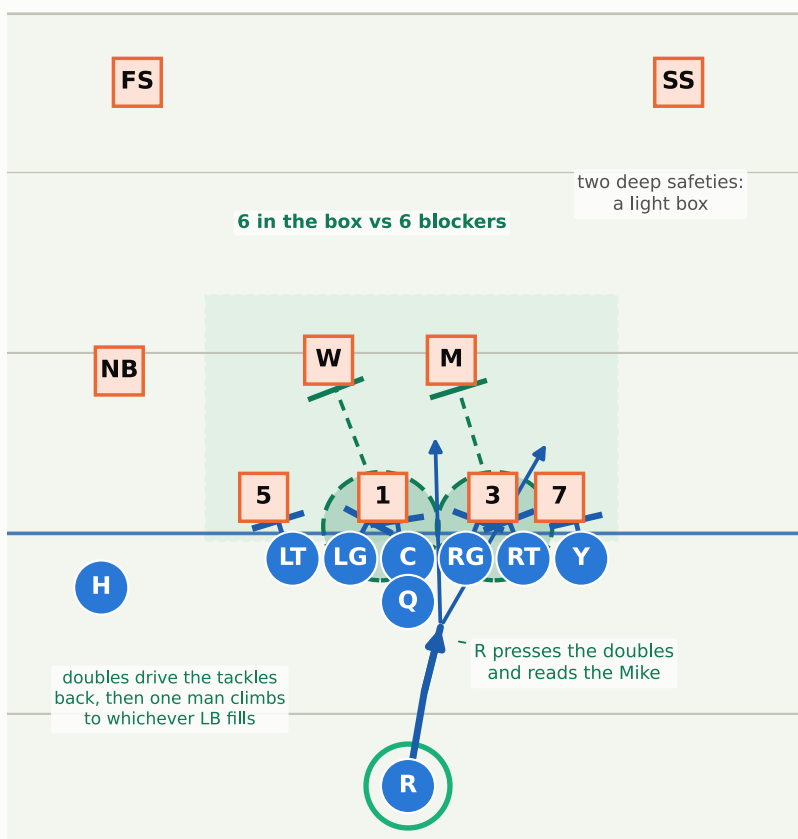


Figure 5: Duo against a two-high nickel defense: six in the box (four linemen and two linebackers; the nickel back is out over the slot) against six blockers (five linemen and the Y). Two double teams (green ovals), the LG and C on the 1 and the RG and RT on the 3, drive the tackles straight back; the LT and the Y block the ends man-on. Nobody is assigned to the linebackers: the back presses the double teams and reads the Mike, and whichever linebacker fills, one man comes off the nearest double team to block him (green dashes). The back hits the gap the Mike leaves: the A gap if the Mike fills outside, or the B gap if he fills inside. Two deep safeties are 12 yards away; the receivers and corners are outside the picture.

Where do duo's extra blockers come from, if nobody pulls? From the **defense's own choice**. Look at the box in Figure 5: six defenders against six blockers, with two safeties sitting 12 yards deep. Against a defense that keeps two safeties high to protect against the pass (a light box), the offense already has a hat for every hat, and it does not need to manufacture one. What it needs is *movement*, someone driven backward so the back can fall forward for four yards, and two double teams on the two biggest defenders are the most direct way to get it. Duo does not fool anyone. It is a bet that your five linemen and a tight end can move their six, and that your running back can make the last man miss.

The running back's job is the subtle part. Since the linemen block the defensive linemen first and climb late, the linebackers are, for a beat, unblocked, and the back is the one who "blocks" them with his path. He presses the double teams, aiming at the gap between them, and watches the Mike. If the Mike fills one gap, the nearest double team sends a man up to him and the back takes the other. It is a read, like zone's, but a vertical one: the back is choosing which side of a double team to run past, not whether to cut back across the field.

Duo's trade is plain. It needs blockers who can move people straight back, and it does nothing to the numbers: against a seventh or eighth defender in the box, there is nobody to block him. That makes duo a **two-high play**, and its rise in the last decade tracks the rise of two-high defenses: when defenses take a safety out of the box to stop the pass, offenses answer by running straight at the six who are left. NFL Next Gen Stats measured split-safety coverage on 42% of plays in 2025, the most since its tracking began in 2018, which tells you how many light boxes NFL offenses now see.⁴

⁴NFL Next Gen Stats: split-safety coverage on 42.0% of plays in 2025, the highest since tracking began in 2018 (32.9%): <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm>.

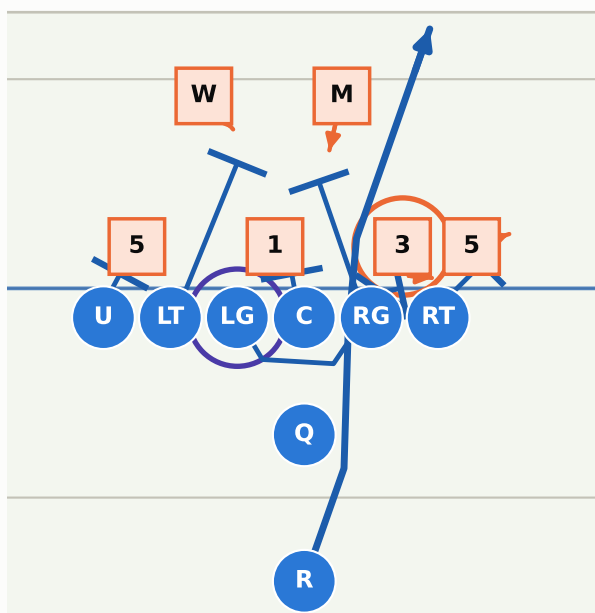
The season: Derrick Henry carried 378 times for 2,027 yards and 17 touchdowns, all league highs, and became the eighth player to run for 2,000 yards in a season. The offense, coordinated by Arthur Smith, mixed outside zone with a heavy diet of duo, often from two- and three-tight-end sets.⁵

What to notice: Henry is listed at 6-foot-2 or 6-foot-3 and about 250 pounds, and, by every account, at his best running *downhill*, building speed straight at the defense rather than stopping and starting. Duo gives him exactly that: on the snap, watch the two double teams surge forward, then watch Henry press between them, find the linebacker, and run through the gap the linebacker did not fill. Count how often his first contact comes four or five yards past the line.

Trap: let him in, then hit him from the side

Trap is one of the oldest plays in football and still one of the most satisfying to watch, because it uses a defender's aggression against him. The offense leaves a defensive lineman *unblocked* at the snap. He feels nobody in front of him and does what he is paid to do: he charges across the line. Then a pulling lineman, usually the backside guard on a short pull, hits him from the side, where he is not looking, and the ball carrier cuts just inside him.

The assignments



1.4 s after the snap

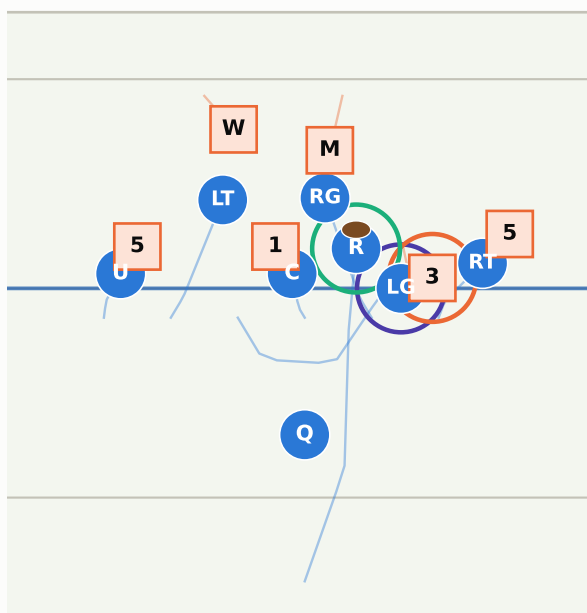


Figure 6: Trap right against a 3-technique, from the pistol. Left, the assignments: nobody blocks the 3-technique (orange ring) at the snap; the right guard releases inside him to the Mike, the left tackle releases inside to the Will, the center blocks back on the 1, the right tackle blocks the 5, and the U blocks down on the backside end. The left guard (purple ring) pulls a short distance behind the center and traps the 3 from the inside. Right, the same play simulated, about 1.5 seconds after the snap: the 3 has come across the line and been kicked out toward the sideline, the linemen are on the linebackers, and the back (green ring) is through the hole inside the trap.

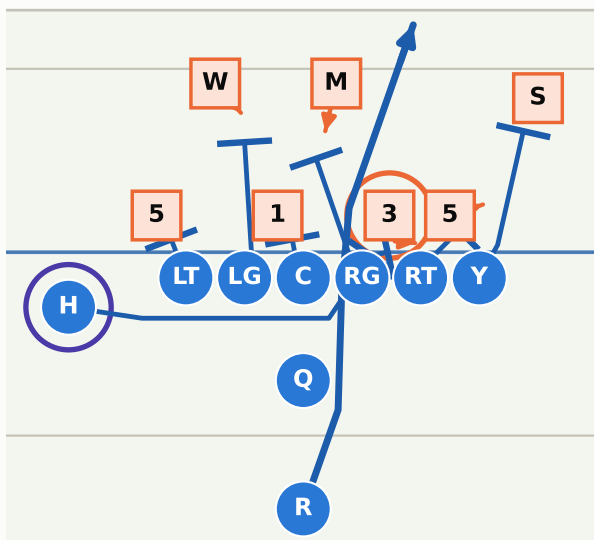
Why would anyone *invite* a defender into the backfield? Because of what it does to the rest of the line. With the 3-technique left alone, the right guard does not have to block him and can go straight to the Mike, and the left tackle can go straight to the Will: in Figure 6 both linebackers are blocked within a second, which is hard to do on most runs. And the defender who was left alone

⁵Henry's 2020 season ("a league-high 378 times for a league-high 2,027 yards and a league-high 17 touchdowns"): CBS Sports, <https://www.cbssports.com/nfl/news/falcons-coach-arthur-smith-well-use-multiple-backs-trust-scheme-to-replicate-derrick-henrys-success>; the eighth 2,000-yard season in NFL history: Tennessee Titans, <https://www.tennesseetitans.com/news/dh2k-titans-rb-derrick-henry-eclipses-2-000-yard-mark-in-win-over-texans>. Henry's listed height differs by source (6-2 on the Ravens' roster, 6-3 elsewhere). Tennessee's duo run game ("multiple double-teams along the interior," the line working "directly upfield"), Arthur Smith's system from 2019, and the Titans leading the league in 13 personnel in 2019: Ben Solak, "Without Derrick Henry, the Titans Offense Is Losing Its Identity," The Ringer (Nov 19, 2021), <https://theringer.com/2021/11/19/nfl/titans-offense-derrick-henry-identity>.

Trap is a **counter-punch to penetration**. It is most dangerous against an aggressive, one-gap defensive tackle, the kind of 3-technique who lives in the backfield, and against defenses that slant hard at the snap. It is least dangerous against a disciplined lineman who reads the guard: if the guard in front of him releases *inside* instead of blocking him, a well-coached tackle knows the trap is coming, squeezes down, and closes the hole before the trapper gets there. Coaches answer with constraint plays such as the *influence* trap, in which the lineman in front of the trapped man pretends to pass-block, inviting the defender upfield, before the trap hits him.

Wham: the trap a tight end throws

The assignments



Swapping a guard for a tight end changes the arithmetic in the offense's favour. On a trap, the guard who pulls is one of the five men who would otherwise block somebody; on a wham, the trapper is a tight end who would otherwise be blocking an edge defender or running a route, so the whole interior line is free. In Figure 7, five linemen and a tight end handle the 1, both ends and all three linebackers, and the H takes the defensive tackle. The obvious objection is size: a 250-pound tight end against a 300-pound tackle. The answer is the angle. The tackle is charging upfield, looking at the quarterback, and the tight end has a running start into his hip. It is, as Chris B. Brown put it, a surprisingly simple block, because the defender is so focused on getting into the backfield that he never sees it coming.⁷

⁷Chris B. Brown, "Draw It Up: The 49ers and the 'Wham' Play," *Grantland* (Oct 17, 2011): the wham is "similar to a traditional trap play, except the trap block comes from the H-back, not a guard," and is "a surprisingly simple block": <https://grantland.com/the-triangle/draw-it-up-the-49ers-and-the-wham-play/>.

The play: in Week 6 of the 2011 season at Detroit, Jim Harbaugh's 49ers (offensive coordinator Greg Roman) faced Ndamukong Suh, one of the most destructive interior linemen in football. Their plan was to *not* block him: on wham runs, the line left Suh alone and released to the linebackers, and tight end Delanie Walker came across and hit him from the side as he charged into the backfield, with Frank Gore running past.⁸

What to notice: wham is easiest to spot from the end-zone replay. Look for a defensive tackle who comes free through the line untouched, *too* free, and then a tight end arriving at his hip from the other side of the formation. Then look at the second level: on a well-blocked wham, every linebacker has a lineman on him.

Roman took the same toolkit (power, counter, trap and wham, plus fullbacks and extra tight ends) to Baltimore, where it became half of the most productive rushing offense in NFL history; the film room on the 2019 Ravens is below.

Iso: the fullback against the linebacker

Iso, short for **isolation**, is the simplest run in this chapter. Every lineman blocks the defensive lineman on him, the uncovered lineman takes the backside linebacker, and the fullback runs through the hole and blocks the play-side linebacker: one man, alone, against a fullback with a running start.

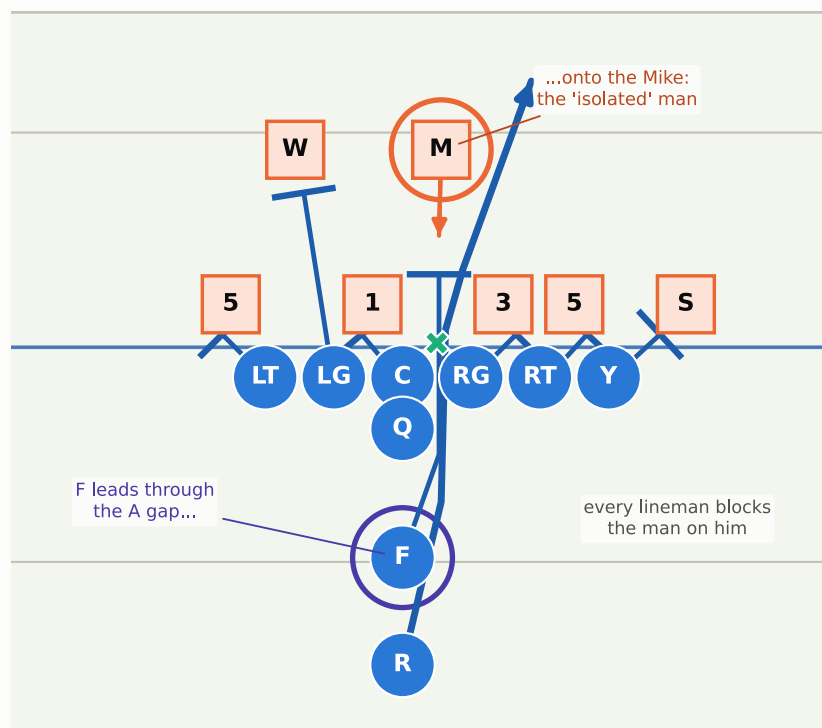


Figure 8: Iso right against the same 4-3 Over: no pullers, no double teams. Each lineman blocks the man on him (the RG turning the 3 out, the center turning the 1 to the left, the tackles and the Y blocking the ends and the Sam), and the uncovered left guard goes to the Will. That leaves the Mike, ringed, 'isolated': the fullback (purple ring) leads through the strong-side A gap (green X) and blocks him as he fills, and the back reads the fullback's block and cuts off it. Two deep safeties, not shown, are the only defenders nobody blocks.

Iso's logic is a clean version of hat math. Against a seven-man box (four linemen, three linebackers), the offense has seven blockers: five linemen, a tight end and a fullback. Every defender gets a hat, and the one matchup the play depends on is the fullback

⁸Same Grantland column (the 49ers' 25–19 win at Detroit on Oct 16, 2011, Week 6; the plan to leave Ndamukong Suh unblocked; Delanie Walker's wham blocks; Frank Gore). On Roman's role in the 49ers' run game: NBC Bay Area, "Roman Is 49ers' Genius Behind Innovative Running Attack": <https://www.nbcbayarea.com/news/local/roman-is-49ers-brain-behind/1954545/>.

against the Mike. Where it hits is decided by the front: iso usually aims at a *bubble*, the gap behind an uncovered lineman, where a linebacker rather than a defensive lineman is the nearest defender, so the fullback can get to him.

The trade is that iso gains nothing in the numbers and moves nobody sideways; it is a straight contest. Its virtues are speed and simplicity: the ball hits fast, and the back has one read (which side of the fullback's block to take). Iso was a staple of the I-formation era, and it has faded with the fullback. In 2025 only about one NFL snap in ten came from personnel with two backs (a fullback or a second running back), so most teams now get the same effect by leading a tight end or H-back through the hole, which is zone insert by another name.⁹

Toss, sweep and pin-and-pull: power goes outside

Everything so far hit between the tackles or just outside them. The gap family also has outside runs, and they are built from the same parts: block down, pull, lead.

A **toss** is the quarterback pitching the ball underhand to a back who is already moving toward the sideline; a **sweep** is any outside run with blockers leading the back around the corner, whether the ball is pitched or handed off. The toss saves time: the back catches the ball two or three yards wider and already at speed, instead of waiting for a handoff and then turning the corner.

The ancestor: Lombardi's sweep

The most famous sweep in football is the **Packers sweep** of Vince Lombardi's Green Bay teams of the 1960s.

The Packers sweep (1960s)

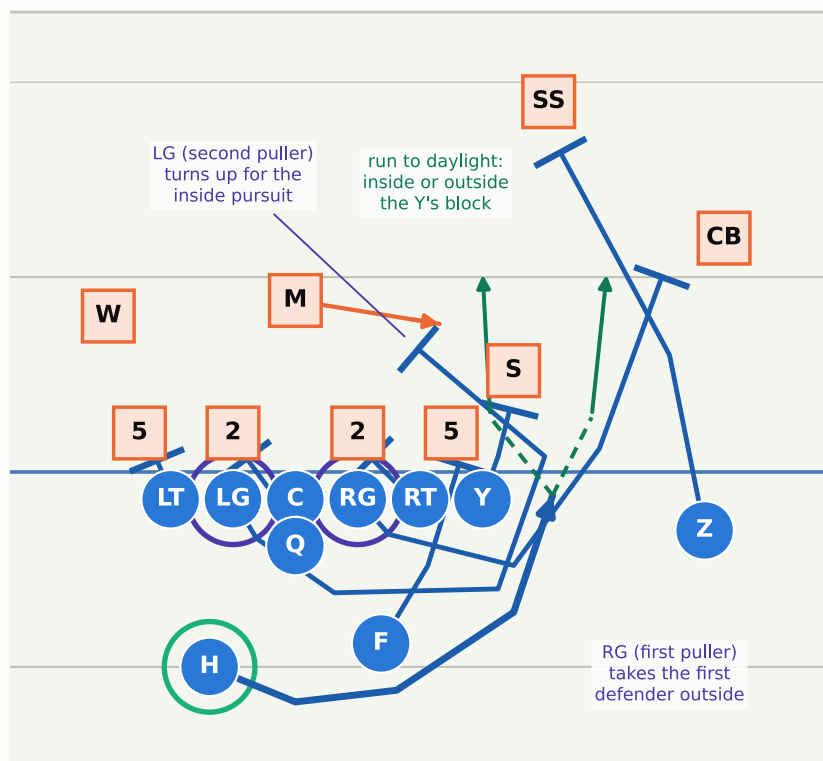


Figure 9: The Packers sweep, as Lombardi's 1960s teams ran it (an illustrative reconstruction against a 4-3 of that era; the back's exact landmarks varied). Both guards pull (purple rings): the right guard, first, takes the first defender to show outside (here the cornerback); the left guard, second, turns up inside for the pursuit (here the Mike). The tight end blocks the Sam, the right tackle blocks down on the tackle over the pulling guard, the center blocks back, and the fullback blocks the end. The halfback (green ring) reads the tight end's block and runs to daylight: inside it if the Sam is driven outside, outside it if the Sam fights inside.

Guards Jerry Kramer and Fuzzy Thurston pulled in tandem, leading Paul Hornung or Jim Taylor around the end, and the play was the cornerstone of an offense that won five NFL championships in seven seasons (1961, 1962, 1965, 1966 and 1967, the last two

⁹This book's calculation from nflverse play-by-play and participation data (`load_participation`), 2025 regular season, pass and run plays with a down: 10.3% of 32,813 plays had two or more running backs and fullbacks on the field (21 personnel alone was 7.0%). Reproduced in nflverse in Python.

including Super Bowls I and II).¹⁰ Lombardi's own phrase for the back's job became the title of his 1963 book, *Run to Daylight*: the back did not run to a spot drawn on the card; he read the tight end's block on the outside linebacker and went wherever the defender was not. Lombardi drilled the sweep endlessly precisely because everyone knew it was coming, and the answer to "everyone knows" was perfect execution plus a back who could choose his hole late.

Look at Figure 9 with this chapter's vocabulary and you can see it is power stretched sideways: down blocks inside, pullers leading, a fullback on the end, the backside left to chase. The modern descendant is pin-and-pull.

Pin-and-pull

Outside zone (03-02) has a problem against **wide fronts**, defenses whose ends line up well outside the tackles and tight ends. To run outside zone, the play-side blockers must *reach* the defenders in front of them, getting their heads to the outside, and a defender lined up two yards outside you is almost impossible to reach. Defenses that play wide (a wide-9 end, an end in a wide 5, an outside linebacker walked out over the tight end) are partly built to kill the reach.

Pin-and-pull gives up on reaching. Each play-side blocker asks one question about the gap on his *backside*, the side away from the play:

- **Is a defender in it (or shaded into it)? Pin him.** Block down on him, which is an easy angle because he is inside you.
- **Is it empty? Pull.** Run around the blocks outside you and lead.

The beauty of the rule is that it turns the defense's alignment into the offense's advantage. The wider the defenders line up, the more of them end up *inside* an offensive lineman, where he can pin them, and the more linemen are left uncovered, free to pull. Every block in Figure 10 is either a down block or a block by a puller with a running start, and nobody has to reach anyone. The rules adjust to the front without a new call, the way zone rules do, which is why some coaches describe pin-and-pull as a gap scheme that thinks like zone.¹¹

¹⁰"Sweep (American football)," Wikipedia (the Packers sweep; Kramer and Thurston pulling; Hornung and Taylor; "an offense that won five NFL titles in seven years"): [https://en.wikipedia.org/wiki/Sweep_\(American_football\)](https://en.wikipedia.org/wiki/Sweep_(American_football)); Green Bay Packers, "Fuzzy Thurston a guardian angel on famous power sweep": <https://www.packers.com/news/fuzzy-thurston-a-guardian-angel-on-famous-power-sweep>. Vince Lombardi with W. C. Heinz, *Run to Daylight!* (Prentice-Hall, 1963).

¹¹The covered/uncovered pin-and-pull rule ("If there is a defender in their gap they are going to block down on this defender. If there is not a defender ... they are going to pull"): Youth Football Online, "Simple Rules for the Pin and Pull Play": <https://youthfootballonline.com/simple-rules-for-the-pin-and-pull-play>; Chris Haddad, "How To Install The Pin & Pull Play," Viqtory Sports (covered linemen "gap block in the back-side gap," uncovered linemen "pull to the play-side"): <https://www.viqtorysports.com/pin-pull-what-is-it-and-how-is-it-run/>.

Pin-and-pull toss right vs a wide front

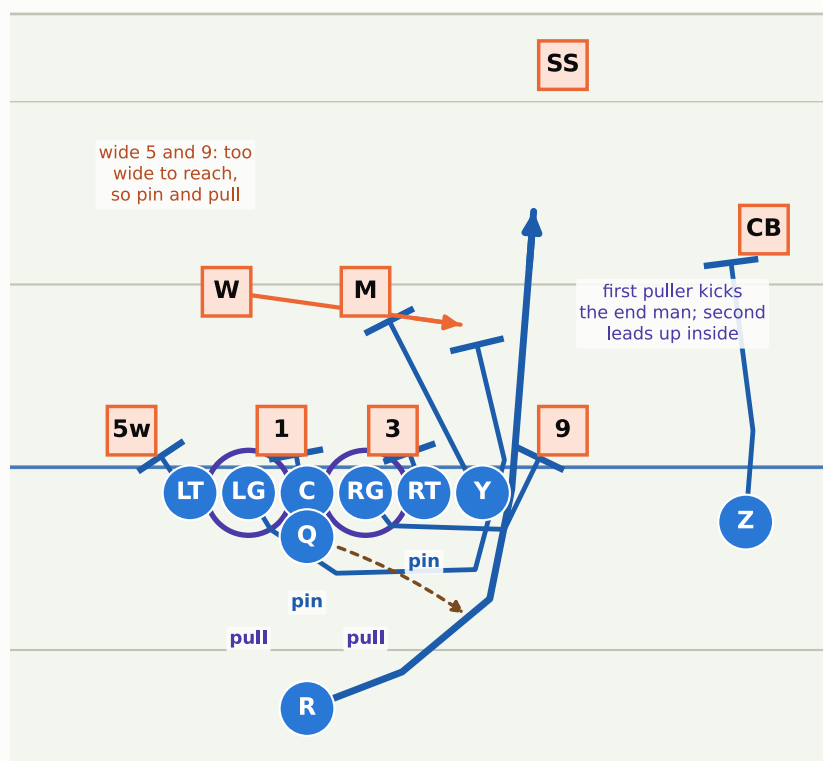


Figure 10: Pin-and-pull toss right against a wide front (an end in a wide 5 on the left, a wide-9 end on the right). The center has the 1 shaded on his backside, so he pins him; the right tackle has the 3 in his backside gap, so he pins him; both guards have empty backside gaps, so both pull (purple rings). The first puller kicks out the wide 9 from the inside, and the second leads up inside him for the Will, who is flowing over the top. With nobody to pin, the Y goes inside to the Mike. The back takes the toss (dashed brown line) and turns up inside the kick-out.

The 2024 Washington Commanders, with Kliff Kingsbury calling plays for rookie Jayden Daniels, leaned on pin-and-pull as the gap-scheme partner to their outside zone; one film breakdown summarized their rule against a front with a 1-technique as “pull both guards: first guard kicks, second guard leads,” which is exactly Figure 10.¹²

The draw: a run disguised as a pass

The last member of the family barely looks like a run at all. On a **draw play** (or simply a **draw**), the offensive line sets up as if to pass-protect, the quarterback shows a drop, the pass rushers are allowed to run upfield, and the ball is handed to the back late, after the linebackers have turned their backs to run to their pass-coverage zones. The rush has *drawn* the defenders out of the play.

¹²Alert the Post, “Cutup Friday: 2024 Commanders Pin Pull RPO’s” (May 16, 2025): “Pin Pull is a gap compliment for Outside Zone”; “Against Over fronts with a 1-technique nose, the Commanders will pull both guards most of the time. First guard kicks, second guard leads”: <https://alertthepost.substack.com/p/cutup-friday-2024-commanders-pin>. Kingsbury was the Commanders’ offensive coordinator in 2024; for 2026 he is the Rams’ assistant head coach (planning/FACTS-current.md §9).

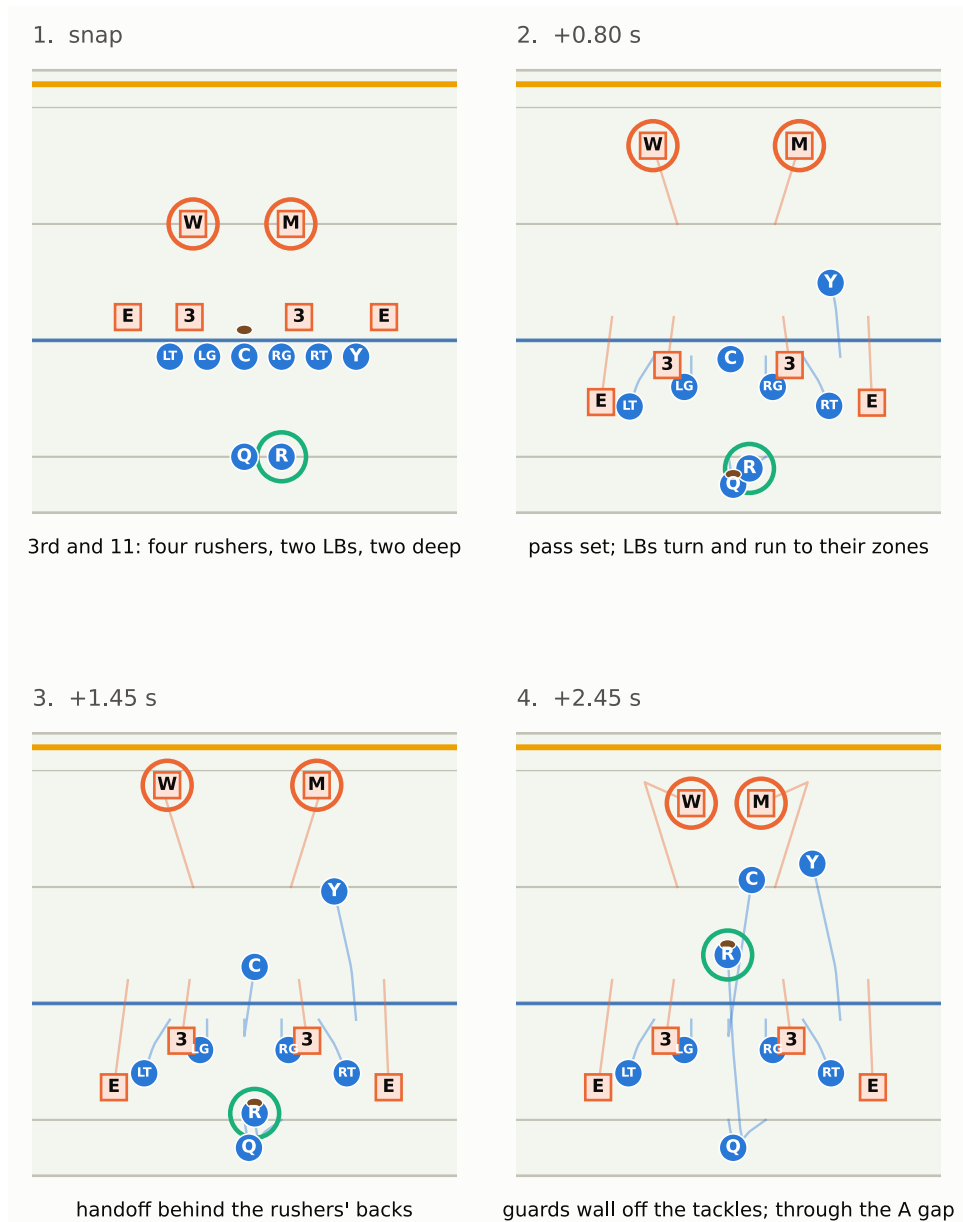


Figure 11: A shotgun draw on third-and-11 against a four-man rush (simulated play; the yellow line is the line to gain). 1: four rushers and two linebackers (orange rings); two safeties are 13 yards deep, above the picture. 2: the line sets as if to pass; the tackles let the ends rush wide and deep, the guards let the 3-techniques come upfield, and the linebackers turn and run to their zones. 3: the quarterback, having shown a drop, hands to the back behind the rushers' backs. 4: the guards wall the tackles off to the outside, the center has released to the second level, and the back (green ring) runs through the A gap toward linebackers who are now nine yards deep and moving the wrong way.

The draw belongs in a gap chapter because it is built on the same principle as the trap: **let the defender go where he wants to go, then block him there**. The guards in Figure 11 do not try to stop the 3-techniques from rushing; they let them come upfield and then turn them outward, away from the A gaps. The tackles ride the ends wide and deep, the way they would on any pass. The more eagerly the defense rushes the passer, the bigger the hole in the middle gets. And the linebackers, who on a normal run would be the problem, have spent the first second running *away* from the ball to cover receivers.

That tells you when to expect it: on **passing downs, against a four-man rush with linebackers who drop**. Third-and-long is the classic draw situation, and a defense with two deep safeties and two linebackers dropping into zones is the classic draw defense. A draw on third-and-11 may not get a first down, but it can steal five or six yards, flip the field position for a punt, and teach the pass rush to be a half-step more careful next time. That half-step is its own reward.

The trade: the draw is helpless against the blitz. If a linebacker comes *forward* at the snap instead of dropping, he is standing in the A gap the play needs, and nobody is assigned to him because the line is busy pretending to pass-block. A defense that sits a linebacker in the middle as a spy on a running quarterback ruins it for the same reason.

Paul Brown's Browns made the draw famous in the late 1940s with Marion Motley, a fullback of about 232 to 238 pounds who could also pass-block, which made the fake believable. The best-known story says the play was discovered by accident: on a pass play gone wrong, Otto Graham improvised a handoff to Motley, who had expected to block, and the rushers ran right past him. Another account has Brown already experimenting with a delayed handoff to Motley at the Great Lakes Naval Station in 1945. Accounts of where and when differ, but the play was a Cleveland staple by the time the Browns joined the NFL in 1950.¹³ Its descendants, the quarterback draw and the run-pass option, belong to Option Football and RPOs.

Gap or zone? How coaches choose

Almost every NFL team runs both families, so the real question is when each one is the better call. Five things push a coach one way or the other.

The front. Gap schemes need *angles*, so they like fronts that give them: a 3-technique to block down on, a 1-technique to block back on, an end man who can be kicked out. They are vulnerable to movement, since a defender who slants into the space a puller is running through can blow up two blockers at once. Zone, built to absorb movement, is the natural answer to slanting fronts. Wide fronts, as you saw, push teams toward pin-and-pull and kick-outs; a penetrating 3-technique invites the trap or the wham; linebackers who flow fast to stop outside zone invite counter.

The defense's shell. Against a light box (two deep safeties, six in the box), the offense already has its hats and wants movement: duo and inside zone. Against a loaded box (a safety down, seven or eight in it), the offense needs a man, and power and counter manufacture one; or it stops running and throws, which is the point of loading the box.

The personnel. Power wants a kick-out blocker: a fullback, an H-back, a second tight end. Iso needs a fullback. Wham needs a tight end who can trap. A team with no fullback on the roster runs a different gap game from one that has an All-Pro at the position.

The offensive line. Zone wants linemen who can move laterally and stay on their feet, and has made a home for lighter, quicker linemen. Gap schemes want linemen who can drive a man off the ball on a down block and guards (or tackles) who can pull and hit on the move. Teams draft and coach their line for the run game they want, and their run game also bends toward the line they have.

The situation and the sequence. Short yardage favours the known hole and the extra hat of a gap scheme. Long yardage favours the draw. And every run sets up another: counter exists because power exists, the trap because 3-techniques penetrate, the bootleg because the backside end chases (Constraint Theory and Sequencing is the full version of that idea).

¹³"Draw play," Wikipedia (the botched pass play on which Otto Graham improvised a handoff to Motley, after which Brown "began to work it in as a regular play"): https://en.wikipedia.org/wiki/Draw_play; "Marion Motley," Wikipedia (at Great Lakes in 1945, "Brown's experimentation with a new play: a delayed handoff later called the draw play"; "listed at 238 pounds"): https://en.wikipedia.org/wiki/Marion_Motley; Pro Football Hall of Fame biography (6-1, 232 pounds): <https://www.profootballhof.com/players/marion-motley/biography/>.

Table 1: Gap and zone, side by side.

	Gap schemes	Zone schemes
Blockers are assigned...	men and gaps, before the snap	areas, by rule after the snap
The hole is...	chosen in advance (it can move a yard: kick or log)	wherever the wall cracks
Extra blockers come from...	pullers, fullbacks, tight ends moved across	double teams; a quarterback who reads the backside end
The back's job	follow the puller, read one block	press the aiming point, read the defense
Beats...	big, static fronts; loaded boxes (with numbers); aggressive penetrators (trap)	slants and stunts; fast-flowing linebackers (cutback)
Struggles against...	movement into the pullers' path; defenders who read the guards	penetration; wide fronts (outside zone)
Tell on the broadcast	a lineman pulls	the whole line steps the same way

In 2019 Kyle Shanahan, the most famous wide-zone coach in football, surprised people by adding more power and counter to his 49ers' run game; he told his team he had called such plays only about seven times in more than a decade as a play-caller.¹⁴ That is the modern consensus in one decision: a scheme is a menu, not an identity.

i Madden vs. real life

In the game, "HB Power" is a play in the menu that either works or doesn't, and the defense picks a play without seeing yours. In a real NFL game the line coach has decided during the week which gap runs fit this opponent's front, the play-caller sequences them so that each one sets up the next (power, then counter off the same look; trap after the 3-technique has been penetrating all half), and the linemen still adjust after the snap: the center makes a call on the front, the puller decides kick or log, the back decides which side of a double team to take. There is no best run, only a best run *against this on this* down.

i Film room: Baltimore Ravens 2019

The season: the Ravens ran for 3,296 yards, breaking the NFL single-season team record the 1978 Patriots had held for 41 years, and Lamar Jackson ran for 1,206 of them.¹⁵ Coordinator Greg Roman's menu mixed nearly every play in this chapter (power, counter, trap and wham, pin-and-pull) with outside zone and Jackson's option reads, often from formations with an extra tight end or fullback Patrick Ricard, and used motion and a changing mix of personnel groups to keep defenses from settling into the right numbers.¹⁶

What to notice: the quarterback is part of the blocking arithmetic. On a Ravens power play, the end left unblocked on the backside is often *read* by Jackson instead of ignored, so he cannot chase the back without giving Jackson the edge. That is the gap scheme's price (the free backside defender) being paid by the option, which is the subject of the next chapter. Roman left Baltimore after the 2022 season and John Harbaugh was fired after 2025, so treat this as a closed case study, told in full in The Ravens' Lamar Jackson Run Game.

¹⁴Nick Wagoner, ESPN, "49ers, Ravens use highly effective, but very different, rushing attacks to excel" (Nov 29, 2019): "the Niners have incorporated more power (or gap) blocking into the mix, something that was previously unheard of for a Shanahan offense"; Shanahan told his team he had used such plays about seven times in his play-calling career before 2019: https://www.espn.com/nfl/story/_/id/28178795/49ers-ravens-use-highly-effective-very-different-rushing-attacks-excel.

¹⁵"Ravens break single-season team rushing record," NFL.com (Dec 2019; 3,296 yards for the season, breaking the 1978 Patriots' 3,165): <https://www.nfl.com/news/ravens-break-single-season-team-rushing-record-0ap3000001092761>; Jackson's 1,206 yards: https://en.wikipedia.org/wiki/2019_Baltimore_Ravens_season. Harbaugh's firing in January 2026: [planning/FACTS-current.md](https://en.wikipedia.org/wiki/2019_Baltimore_Ravens_season) §0 (C3).

¹⁶USA Football, "How Greg Roman's schemes revitalized the Baltimore Ravens' rushing attack" (power, outside zone, pin-pull and counter/trap/wham as the core, with tight ends and fullbacks as extra blockers): <https://blogs.usafootball.com/blog/6397/how-greg-roman-s-schemes-revitalized-the-baltimore-ravens-rushing-attack>. The Ravens "keep defenders thinking with their array of motion and personnel groupings": Wagoner, ESPN (Nov 29, 2019), cited above.

The season: Saquon Barkley ran for 2,005 yards in 16 games, the ninth 2,000-yard season in NFL history, and the Eagles beat Kansas City 40–22 in Super Bowl LIX. A *Philadelphia Inquirer* film study that November described an offense that alternated zone and gap runs “noticeably,” using gap-scheme changeups with pulling linemen to balance its zone game, behind a line coached by Jeff Stoutland.¹⁷

What to notice: pick one Eagles drive and, before every run, guess zone or gap from the formation; then, on the snap, check your guess against the line: does everyone step the same way, or does somebody pull? Then watch Barkley on the gap runs. A back with his speed can follow a puller *patiently*, a half-step behind, and still outrun the defense to the edge once the kick-out lands. (Kellen Moore, the 2024 offensive coordinator, left to become the Saints’ head coach; the Eagles’ line and staff have turned over since, as *The Eagles: Trenches* and *the Tush Push* explains.)

Spotting a puller before the snap

Because a puller runs to the hole, finding him early is the single most useful habit in watching gap runs. Defenses know this too: many linebackers are taught to read the guards, and a pulling guard is their signal to run with him. So both sides are looking for the same tells, and offenses work to hide them.

Before the snap, look for:

- **The backfield.** A fullback or H-back offset to one side is often the kick-out man, and the play usually goes *toward* him on power. A back set deeper than usual needs time, which counter and the sweep require (the pullers have to get there first). In the shotgun, gap runs, like zone runs, usually send the back across the quarterback’s face, so a back offset to the left usually runs right; the puller’s direction confirms it.
- **The guards’ stance.** A lineman about to pull often has less weight on his hand, his feet a little more staggered, his hips ready to turn. This is the heavy hand tell in reverse: a puller is going *sideways*, not forward, and many give it away with a light hand or a slight cheat backward off the ball, since depth gives him room to clear his teammates’ heels. The best pullers hide it, and some offenses deliberately line up their guards the same way on every snap.
- **The formation and the tendency.** Certain personnel groups and formations are gap-heavy for every team (two tight ends with one off the ball, the I-formation, an H-back on the wing); the broadcast graphics and your prediction log will teach you which.

At the snap, the first step settles it. A pulling lineman’s first step is backward and sideways, opening his hips toward the play (some coaches call it a drop step); a zone lineman’s first step is sideways and forward; a pass-blocking lineman’s first step is straight back with his shoulders still square to the defense. One look at the backside guard tells you which.

Offenses exploit the defenders who read guards. A pulling guard on a play-action pass pulls linebackers toward the fake, and a “false pull” on a run sends them the wrong way; counter itself is a lie told with the back’s first step. Those plays exist precisely *because* the tell is so reliable. Trust the guard, but notice when the offense is using your trust against you.

¹⁷Barkley, 2,005 yards in 16 games, the ninth 2,000-yard season: NFL.com, <https://www.nfl.com/news/saquon-barkley-becomes-ninth-player-to-rush-for-2000-yards-in-a-season>; Pro Football Reference, <https://www.pro-football-reference.com/players/B/BarkSa00.htm>. EJ Smith, “Eagles film review: How Saquon Barkley became the NFL’s most explosive running back behind Birds’ O-line,” *Philadelphia Inquirer* (Nov 6, 2024; “the Eagles alternate between the two noticeably”; “the gap-scheme changeups the Eagles are so adept at calling to balance out the zone runs”): <https://www.inquirer.com/eagles/saquon-barkley-explosive-eagles-runs-offensive-line-film-room-20241106.html>. Kellen Moore, the Eagles’ 2024 offensive coordinator, was hired as the Saints’ head coach on Feb 11, 2025: <https://www.inquirer.com/eagles/eagles-kellen-moore-hired-saints-coach-20250211.html>. Super Bowl LIX, Eagles 40, Chiefs 22, Stoutland as the Eagles’ line coach and his departure after the 2025 season: *planning/FACTS-current.md* §1, §9.

Watch for it on Sunday

Watch for it

Follow the guard. For one quarter, on every run, watch the *backside* guard (the one away from where the back lines up, or, if you cannot tell, the left guard) instead of the ball.

Before the snap

- Count the box and find the extra blockers: a fullback, an H-back, a second tight end. Where would a kick-out come from?
- Check the guards' stance and depth. Is one a little off the ball, weight back?

The first half-second

- **Does a guard pull?** If yes, call "gap" and say the direction out loud: the play is going where he is going.
- **Does a second lineman pull with him?** Two pullers (guard and tackle, or guard and tight end) means counter. A pull with the back stepping the other way also means counter.
- **Nobody pulls, but the line fires straight ahead in double teams?** Duo. **The whole line steps the same way?** Zone.
- **A defensive tackle comes through untouched?** Look for the trap (a guard) or the wham (a tight end) arriving at his hip.
- **The line drops back as if to pass, and the back gets the ball late?** Draw.

After the play (on the replay)

- Find the kick-out. Did the puller kick the end out or log him in, and did the back go the right way off it?
- Find the free defender: on power, the backside end. Did he make the tackle?

In your prediction log, add one letter after each run: **G** (gap) with the play's name if you can (P, C, D, T, W, I, S for power, counter, duo, trap, wham, iso, sweep) or **Z** (zone). After a quarter, look at how often the guard told you the truth.

Predict the play

Predict the play

Drill 1. Follow the guard. First-and-10. Figure 12 shows the first half-second of a run from the I-formation with the tight end on the *left*. Name the play, say where it will hit, and say who will kick out, who will lead through the hole, and which defender nobody is going to block.

Drill 1: the first half-second. What is it?

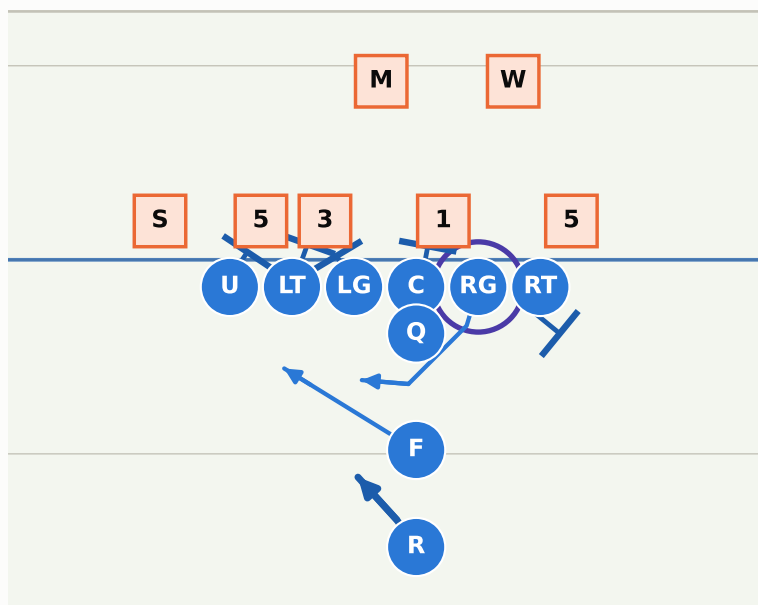


Figure 12: Drill 1: the first half-second. The U tight end (left) and the left tackle have stepped inside, the left guard is stepping toward the 3-technique, the center is stepping right, the right guard (purple ring) is pulling left, the fullback is heading for the left edge, and the running back has taken two steps to his left.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 2. Third-and-long. Third-and-11 from midfield. The offense is in the shotgun with three receivers and a tight end split out; the defense shows a dime package (six defensive backs) with two deep safeties, four linemen in wide alignments, and one linebacker seven yards deep (Figure 13). The offense is going to hand the ball off. Which run from this chapter has the best chance, and what would ruin it?

Drill 2: 3rd and 11. Which run?

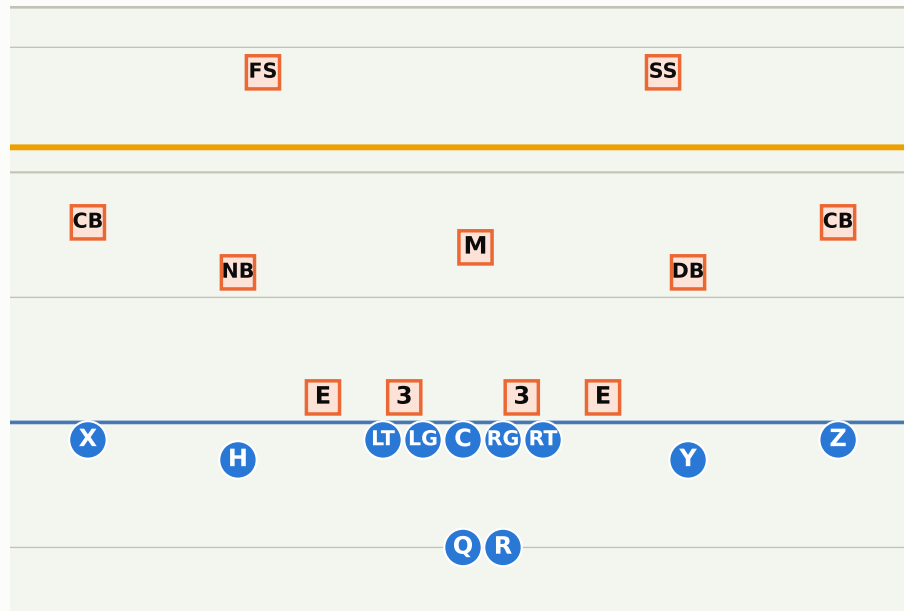


Figure 13: Drill 2: third-and-11 (yellow line). Shotgun, the back beside the quarterback, the tight end (Y) in the slot; a dime defense with four rushers aligned wide, one linebacker (M) about seven yards deep, and two safeties 14 yards deep.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 3. Kick or log? Counter right. Figure 14 freezes the play as the pulling left guard, whose job is the end man, arrives. The end has squeezed hard inside as the tackle blocked down. What should the puller do, where should the back go, and which defender now has to make the tackle?

Drill 3: kick or log?

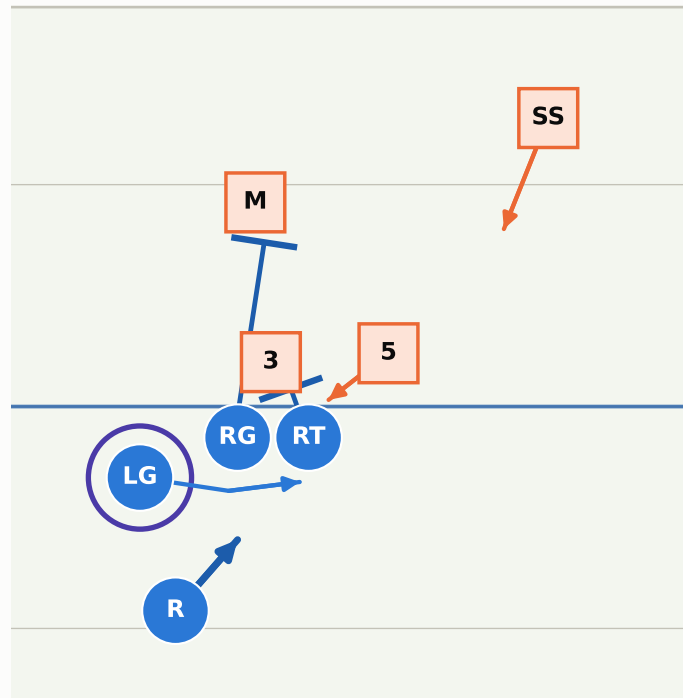


Figure 14: Drill 3: the moment of decision. The right tackle has blocked down on the 3; the end (5) has crashed inside behind him (orange arrow); the pulling left guard (purple ring) and the back are arriving. Outside, the strong safety (SS) is coming down.

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **A gap scheme blocks men, not areas, and brings extra blockers to a hole chosen in advance.** Every gap run is down blocks (the inside wall), a kick-out (the outside wall) and a puller through the middle, and the hole moves only as far as the end man forces it (kick or log).
- **Power is the template.** Block down, kick out with the fullback, lead with the backside guard, hinge the backside, and leave the backside end unblocked. It turns a play side of three blockers against four defenders into five against four.
- **The rest of the family varies one thing.** Counter changes the back's first step and pulls two (guard kicks, tackle wraps). Duo pulls nobody and drives two double teams into a light box. Trap and wham leave a lineman unblocked and hit him from the side, with a guard or a tight end. Iso sends a fullback at one linebacker. Pin-and-pull and the sweep take the same blocks outside. The draw blocks the rush where it wants to go.
- **Every gap run names who is left free.** Usually it is a defender far from the hole (the backside end); the offense bets he arrives late, or reads him with the quarterback.
- **Coaches choose by front, shell, personnel and line.** Gap schemes like angles, loaded boxes and big static fronts; zone likes movement and light boxes; most teams run both from the same looks.
- **Follow the guard.** A puller runs to the hole. Find the backside guard on every run, and you will call the direction of most gap plays before the back gets there.

Connections

Builds on: The Language of the Run Game (down blocks, pulls, kick-outs, lead blocks, hat math) and Zone Running (the other family, and the contrast that runs through this chapter); Classic Formations (the I-formation and the fullback) and How to Watch a Broadcast (stance tells and the prediction log).

Sets up:

- Option Football: the quarterback who *reads* the defender power leaves unblocked, plus QB power and QB counter, where the quarterback carries the ball behind these same blocks.
- Perimeter Runs and Misdirection: crack and seal blocks on the edge, jet sweeps, and the Wing-T's buck sweep, a gap-scheme cousin of the Lombardi sweep.
- Play-Action, Boots and Screens: pulling guards on pass plays, and why linebackers who read guards are vulnerable to them.
- Run Fits and Box Math: the defense's half of this chapter: who fills which gap against power, and the spill-or-box decision that decides kick or log.
- Constraint Theory and Sequencing: power, counter, trap and the boot as a sequence in which each play punishes the defense's answer to the last.

See also (case studies): The Ravens' Lamar Jackson Run Game and The Eagles: Trenches and the Tush Push.

If you want to go deeper: Chris B. Brown's *The Art of Smart Football* and his Grantland "Draw It Up" columns (the wham piece cited here is a model of the form); Vince Lombardi and W. C. Heinz, *Run to Daylight!* (1963), for the sweep in its inventor's words; and any coaches' clinic talk on "gap, down, backer" rules, which are the grammar of every play in this chapter.

Go deeper

Finding gap runs in real tracking data. The play diagrams and both animations here are gridiron simulations written in NFL Big Data Bowl coordinates. The 2025 Big Data Bowl release (2022 season, weeks 1–9) labels every run with PFF's primary run concept, so you can pull every power or counter play, find the pulling guard by his first-second movement, and draw the real version of Figure 3. The data needs a free Kaggle login, so this code (in the web edition, under "Show the Python") is not run when the book is built.

Answers to Predict the play

Answer 1.

Power left. It is Figure 1 in a mirror. The play side is the *left*: the U tight end and the left tackle are blocking down, the left guard is joining the left tackle on the 3-technique (he will climb to the backside linebacker, the Will, now on the right), and the center is blocking back on the 1. The **right guard is the puller**: he will turn up through the hole just outside the U's down block and lead on the play-side linebacker (the Mike). The **fullback kicks out** the end man, the Sam standing up outside the U. The hole is off-tackle on the left, between the U's down block and the kick-out.

Nobody blocks the backside end, the 5-technique on the right tackle, who is hinging. That is power's price.

The tells you could have used: the back's first steps and the fullback both went left, and one guard pulled left. When the backside guard pulls, the play goes where he goes.

Answer 2.

The draw. Everything in the picture is a draw invitation. The defense has five men in the box against five offensive linemen, the rushers are lined up wide so they can get to the quarterback, and the only linebacker will drop into a zone at the snap. If the line sets up to pass, the wide ends will rush upfield and outside, the 3-techniques will rush through the B gaps, and the middle of the field, the A gaps, will be empty except for the linebacker, who will be backpedalling. Hand the ball off late, after the rush has committed, and the back is running at a linebacker who is turned the wrong way and safeties 14 yards deep.

Will it get 11 yards? Maybe not, and the offense knows that. A draw here is often field-position football (a five-yard gain makes the punt shorter for the other team) and a warning shot to the pass rush.

What would ruin it: the linebacker coming *forward* at the snap, a blitz up the middle, or a "spy" assigned to sit in the middle and watch the back. Any defender who stays home in the A gaps is standing exactly where the draw needs to go, and nobody is assigned to block him.

Answer 3.

Log him, and the back bounces outside. The end has taken away the inside: there is no room to kick him out, because he is already inside where the hole was supposed to be. So the guard aims for the end's *outside* shoulder, gets his head and body outside him and seals him in (a log block), and the back, reading the guard's helmet, takes the ball outside the log.

The defender who now has to make the tackle is the **strong safety**, coming down outside. That is not an accident: a defense that tells its end to squeeze and "spill" the ball outside is counting on a fast defender outside to meet it there (Run Fits and Box Math explains who that is and why). The offense's counter-move is a block on that man: a wide receiver crack block or a second puller, the subject of Perimeter Runs and Misdirection.